

input $x \in \mathbb{R}$

input $\mathbf{x} \in \mathbb{R}^D$

output $f \in \mathbb{R}$



number (1×1)
derivative $f'(x)$

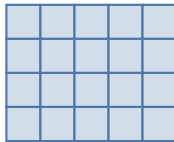


row ($1 \times D$)
gradient ∇f

output $\mathbf{f} \in \mathbb{R}^E$



column ($E \times 1$)



matrix ($E \times D$)
Jacobian J